

SCI501 Final Report: Evaluating unimodal and multimodal approaches to regression task using pretrained embeddings.

Introduction

Multimodal machine learning (MMML) refers to models that extract from multiple data modalities such as text, images, audio and video. This concept has become increasingly important, as real-world information is inherently multimodal, whilst unimodal systems often fail to capture the complexity of these systems. Prior to the rise of deep learning, multimodal systems relied heavily on hand-crafted features and explicit fusion strategies to combine inputs. While effective in constrained settings, these approaches were limited in scalability (Potamianos et al., 2003).

Recent advances in deep learning and transfer learning have substantially improved the ability of models to learn cross-modal representations directly from data and apply them to a range of problems. Large language models (LLMs) have demonstrated remarkable success in natural language processing and this success has driven growing interest in extending these architectures to incorporate vision, audio and video modalities (Xu et al., 2023). This has resulted in multimodal models achieving strong performance on tasks such as visual question answering (VQA), image–text retrieval and document understanding.

More recently, large multimodal language models (MMLLMs) such as GPT-4o, Gemini 1.5, Claude 3.5 and InternVL2.5 have pushed technologies further by integrating modality-specific encoders into a shared transformer backbone (OpenAI, 2024; Google, 2024; Anthropic, 2024; Zhang et al., 2024). By mapping different modalities into a common token space, these systems support unified multimodal reasoning within a single architecture.

Despite these advances, training or fine-tuning large multimodal models remains computationally expensive and impractical for many applied settings. Consequently, pretrained multimodal embeddings are an attractive alternative, offering strong representational power at relatively low computational cost. Understanding how different modalities contribute to predictive performance and how fusion strategies affect learning in various domains is important for understanding these black box systems.

This project attempts to explore the effectiveness of pretrained multimodal embeddings for a real-world regression task: predicting product prices from catalogue text, images and numeric features. By comparing and evaluating unimodal and multimodal models under consistent experimental conditions, this work aims to assess the contribution of each modality and the benefits of late fusion in a practical multimodal learning setting. Through this analysis, the project seeks to confirm concepts of multimodal learning within the e-commerce domain.

Methods

Dataset.

The dataset used for this study was provided from Kaggle as part of a machine learning challenge. The training data is formatted as a CSV containing `sample_id`, `catalog_content`, `image_link` and `price` headers. The objective is to utilise features of an Amazon catalogue (text descriptions, quantities, units and images) to predict the product price. Early exploratory analysis revealed that the target value was extremely right skewed (Figure 1). Several images were unavailable for download and hence the corresponding entries were culled and aligned to account for this.

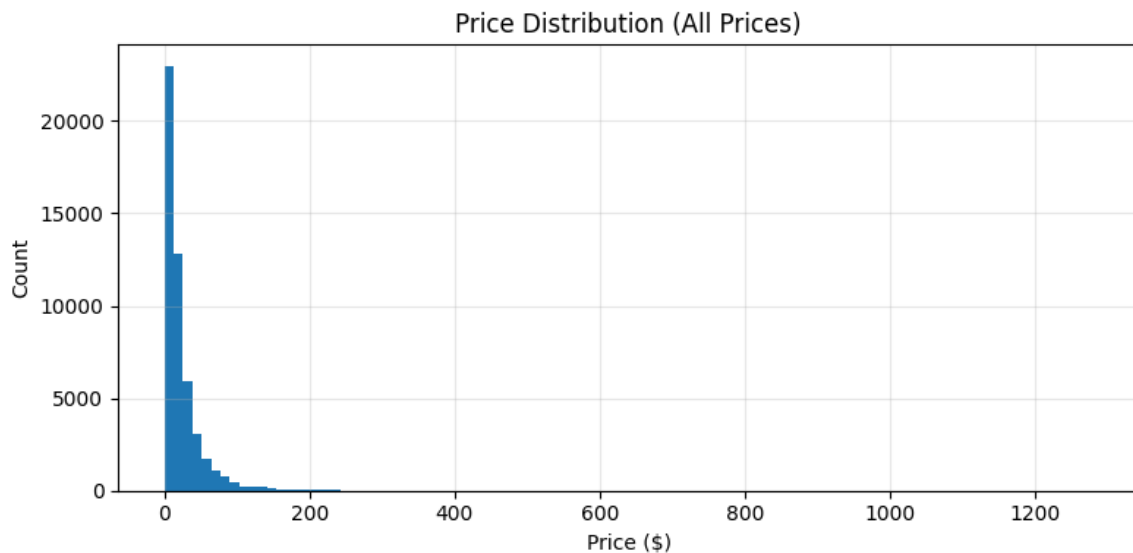


Figure 1. Target price distribution.

Feature extraction and preprocessing.

The raw catalogue content contained a combination of field headings, bullet point paragraphs and unit-value annotations. These aspects add noise to text representations of

our dataset. To combat this, RegEx filtering was applied to strip structural labels and formatting, leaving a clean and consistent product description for each entry (Figure 2).

Gift Basket Village Gourmet Meat and Cheese Gift Set - Swiss Cheese & Summer Sausages - Get Well Soon Gift Baskets for Women - Sympathy Gift Baskets - Perfect for Holidays & Special Occasions Sympathy Gift Baskets: Perfect for sending heartfelt condolences, our gourmet meat and cheese gift baskets include summer sausage and hard salami, offering a comforting selection of charcuterie food items for grieving gifts for women and men.

Figure 2. Processed text sample.

Unit-value pairs were also separately extracted and processed as a numeric feature for the combined model. This action was spurred from the hypothesis that unit-value weights could strongly influence the feature space and thus was given an individual training pathway. Units were provided custom labels based on measurements and then one-hot encoded (Figure 3), whilst values were z-score normalised across splits to prevent data leakage.

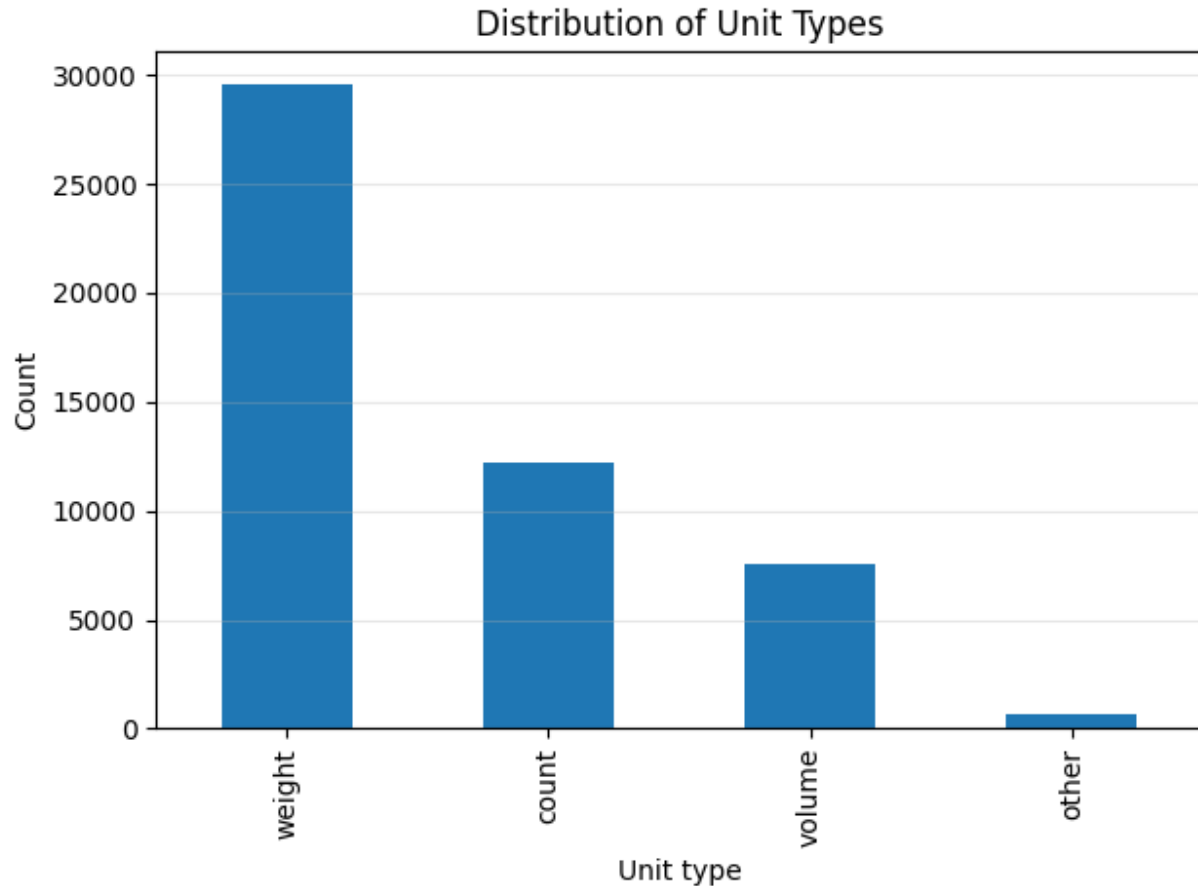


Figure 3. Distribution of one-hot encoded custom labels.

The pretrained sentence embedding model all-MiniLM-L12-v2 was used to convert text into fixed length vectors. Images were encoded using the CLIP ViT-B/32 vision transformer, producing 512-dimensional image embeddings.

Train, Validation and Test Protocol.

Across all experiments, the dataset was partitioned into a 70/15/15 train, test and validation fixed split. The same split indices were reused to ensure maximal fairness in model comparison. Early stopping was implemented during the training and validation steps, whilst the test set was held out for final model evaluations.

To account for the high density of lower cost items, a log transformation was applied to the targets (Figure 4). Value and class based resampling strategies were considered to further address target imbalance, these were ultimately neglected to avoid introducing additional bias and maintain consistency across experiments.

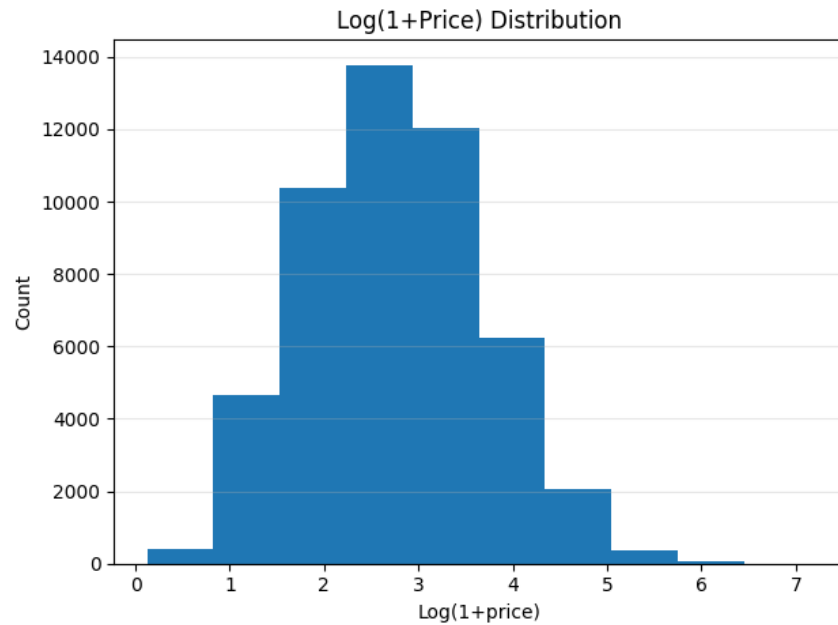


Figure 4. Normalised price distribution using $\log_1 p$ transform.

Model Architecture.

The combined model was a multibranch feedforward neural network regressor in which each modality was transformed independently and then fused together before the final prediction. This process is commonly termed “late fusion”. The text branch takes a dimensional sentence embedding and passes it through a two-layer MLP (linear $\rightarrow$ ReLU $\rightarrow$ dropout, then linear $\rightarrow$ ReLU) to produce a 64-dimensional text representation. In parallel, the numeric branch processes the structured quantity features (scaled magnitude plus one-hot unit encoding) with a single hidden layer (linear $\rightarrow$ ReLU $\rightarrow$ dropout) to produce a 32-dimensional numeric representation. The image branch takes the fixed CLIP image embedding and applies a two-layer MLP (linear $\rightarrow$ ReLU $\rightarrow$ dropout, then linear $\rightarrow$ ReLU) to produce a 64-dimensional visual representation. To stabilise training across different modalities, layer normalisation is applied to the output of each branch prior to fusion. The three normalised vectors are then concatenated ($64 + 32 + 64$) and passed through a shared fusion head consisting of two hidden layers (linear $\rightarrow$ ReLU $\rightarrow$ dropout, linear $\rightarrow$ ReLU $\rightarrow$ dropout) followed by a final linear layer that outputs a single scalar in log-price space (Figure 5).



Figure 5. Combined model architecture.

All experiments were conducted under the same training and evaluation conditions to ensure that differences in performance could be attributed only to information content of the inputs. Each reduced form experiment reused the same split indices, early stopping protocol, optimisation settings, loss function and evaluation metrics as the combined model. Architecturally, each iteration retained the same respective modality specific branch.

Training Procedure.

All models were trained using a custom framework built in PyTorch. Training was performed using the Adam optimiser with a small weight decay term for regularisation and the MSE loss was used as the objective function. Each epoch consisted of a full pass over the training set, followed by evaluation on the validation set with the model dropout disabled and gradients off. Early stopping was applied based on validation loss to mitigate overfitting and the weights with lowest validation loss were retained for final evaluation. After training, predictions were inversely transformed to return to dollar units and performance was reported using MAE and RMSE on the held-out test set.

Results

MAE, RMSE and MAE performance on the test set were measured in US dollars across all experiments (Table 1) as well as true price versus predicted price scatter plots (Appendix 1-3).

Table 1. Error metrics on hold out test set across model runs.

	MSE	RMSE	MAE
Image Model	\$675.36	\$25.99	\$13.20
Text Model	\$698.67	\$26.43	\$13.15
Combined Model	\$642.88	\$25.35	\$12.67

Prices were partitioned into equal sized quantiles and test error predictions of the combined model were evaluated (Table 2).

Table 2. Quantile partitioned predictions by price on combined model.

	Absolute Error (USD)	Percentage Error (%)
Price Category (Quantile)		
<u>Very Low</u>	5.21	184.00
<u>Low</u>	4.10	52.94
<u>Mid</u>	4.49	30.99
<u>High</u>	10.08	38.88
<u>Very High</u>	38.46	54.00

Qualitative analysis was performed on the predictions made by the combined model. The ten worst performing predictions by absolute price error were collated and reviewed.

Discussion

Quantitative Performance Across Modalities

Table 1 summarises test set performance across all evaluated models using mean squared error (MSE), root mean squared error (RMSE) and mean absolute error (MAE) in US dollars. The combined multimodal model achieved the strongest overall performance, with a test RMSE of \$25.35 and MAE of \$12.67, outperforming both the image-only and text-only baselines.

While the magnitude of these improvements is modest, the gains are consistent across all three metrics, indicating a genuine benefit from multimodal fusion rather than modal variation. Relative to the strongest unimodal baseline, the combined model reduced RMSE

by approximately \$0.6–\$1.1 and MAE by \$0.5–\$0.6. This suggests that incorporating multiple modalities improves not only sensitivity to large errors (MSE, RMSE) but also typical prediction accuracy (MAE).

The similarity in performance between the image-only and text-only models suggests that both modalities encode overlapping price information. Visual embeddings may capture cues related to packaging, scale and product quality, while textual embeddings encode semantic descriptors such as quantities, ingredients and product category.

It should be noted that direct performance comparisons between unimodal image and text models may reflect not only the relative informativeness of each modality, but also the strength of the pretrained encoders and the data on which they were trained. As such, unimodal comparisons should be interpreted with caution, as they can confuse modality effects with training biases.

Error Behaviour Across Price Quantiles

To further examine test error behaviour across the price distribution, test predictions from the combined model were evaluated within equally sized price quantiles. This analysis reveals a clear heteroskedastic error pattern, with absolute error increasing substantially as true price increases.

Items in the lower price quantiles exhibit low absolute error (\$4 - \$5) but extremely high percentage error, as expected when denominators are small (Table 2). In contrast, the highest price bucket exhibits mean absolute errors exceeding \$38, confirming that expensive items are systematically underpredicted. These findings indicate that the observed improvements from multimodal fusion are concentrated in the densely populated mid-range of the price distribution and did little to improve predictive accuracy against outliers.

This behaviour is consistent with the training data distribution, which is heavily skewed toward low and mid-priced consumer goods.

Qualitative Error Analysis

To better understand the nature of the largest prediction errors, a qualitative analysis was performed on the ten worst-performing test examples by absolute error. Inspection of these cases reveals several consistent failure patterns.

First, many severe underpredictions involved commercial and industrial equipment, such as the heated condiment dispenser. Although these items visually and semantically resemble common consumer appliances, their pricing reflects durability, power capacity and professional use attributes that are not explicitly represented in either the image or text embeddings. As a result, the model defaulted a safer mid-range price prediction.

Second, bulk and wholesale products are frequently mispriced. Examples include large multipacks of coffee, chocolate confectionery and beverage concentrates. In these cases, unit-value information is either weakly encoded or ambiguous, causing the model to underestimate total value. This limitation highlights a structural weakness in the dataset, rather than a failure of multimodal fusion.

Third, several high-error items correspond to premium food products, such as Wagyu beef and specialty organic goods. These products derive a significant portion of their price from brand reputation, sourcing, or certification, none of which are reliably inferable from visual appearance or generic text descriptions alone. Consequently, the model underpredicts price in these cases.

Implications for Price Modelling Context

The combined quantitative and qualitative analyses indicate that multimodal learning improves average prediction accuracy but does not fundamentally resolve pricing ambiguity for unique items combined with weaker descriptions. Multimodal fusion enhances prediction accuracy when modalities provide complementary signals, but cannot compensate for missing information.

Consideration and reflection.

Whilst the model provides moderate predictive power of price for catalogue items, several avenues of improvement were considered but not pursued due to constraints. The most substantial consideration was the handling of units and values. It was a challenging task to apply class labels to data in a way that added meaningful value to the prediction space. The initial decision was to segregate items based on whether measurements were made in volumes, weights or counts. In hindsight, this was poor practice as the scales of the measurements were not standardised. It was deemed that this approach was harmful to model success and was ultimately neglected.

Another potential scrutiny would be the use of lightweight general models. In a practical setting, fine tuning text and image encoders on task related data would provide richer representation of our domain.

A minor setback to the current experimental design was that the procedure was limited to strictly late fusion - concatenating representations prior to prediction. Alternative fusion strategies such as early fusion and encoder linking (co- and cross-attention) may improve model accuracy in realms where features may be noisy.

For sound experimental consistency, hyperparameters were maintained across ablations. Limited hyperparameter optimisation was performed due to time and scope constraints which may put into question the validity of comparative results. Future iterations of this experiment could strive to improve this and implement a Bayesian search algorithm.

When considering the biases in our approach, we assume that product price can be inferred directly from static catalogue features. In reality, pricing is influenced by external factors such as promotions, supplier agreements and other time-dependent market conditions, none of which were accounted for in this model. Additionally, the dataset exhibits a strong skew toward lower-priced items, which despite log-transformation show evidence of bias against higher-priced products.

A remedy to this problem which was not explored in this project would be to augment catalogue with supplementary data, including brand labels, universal unit-values and a more balanced sample of target prices.

Addressing these challenges would be essential for transitioning from experimental modelling to practical pricing systems. Theoretical use cases for a model such as this could include:

1. Price Quality Control – An approach analogous to anomaly detection, where catalogue items that deviate significantly from predictions can be flagged at mass. A model such as this can reduce the impact of data entry errors or poorly classified items.
2. New Listing Price Estimator - Internal sales teams could expose new products without sales history to the model to receive initial estimations of price.
3. Feature Improvement Analysis - By comparing predictions across different models (text-only, numeric-only, image-only), a potential system can be used to infer which modalities contribute most strongly to pricings for different product types. This insight can inform systems design and business decision making, such as prioritising high-quality images for catalogues or richer text descriptions in categories where visual appearance strongly correlates with price.

Conclusion

This project aimed to assess the effectiveness of varying modalities by using pretrained embeddings to predict prices from catalogue data. During experimentation, the combined multimodal model performed the best – providing improved results over its unimodal counterparts across all metrics. Introspective analysis of the high error cases showed that model struggled with unique items, which was suggest to an issue stemming from skewed and misleading data. These limitations were consistent across experiments and highlight that multimodal fusion alone cannot compensate for critical contextual information.

Overall, the results demonstrate that pretrained multimodal embeddings offer a practical and computationally efficient approach for applied regression tasks. However, improving performance for high-value items will require richer structured features, better unit standardisation and wider representations. This work provides a foundation for future research into more expressive fusion strategies and task specific multimodal modelling for pricing systems.

References

Anthropic. (2024). *Claude 3.5 model card*. Anthropic.

<https://www.anthropic.com>

Chen, Z., Bai, J., Qiao, S., & Wang, X. (2024). *InternVL2.5: A versatile vision-language model for document and chart understanding*. arXiv:2407.02110.

<https://arxiv.org/abs/2407.02110>

Google. (2024). *Gemini 1.5 technical report*. Google DeepMind.

<https://deepmind.google>

Lu, J., Batra, D., Parikh, D., & Lee, S. (2019). ViLBERT: Pretraining task-agnostic visiolinguistic representations for vision-and-language tasks. In *Advances in Neural Information Processing Systems* (pp. 13–26).

<https://arxiv.org/abs/1908.02265>

OpenAI. (2024). *GPT-4o system card*. OpenAI.

<https://openai.com>

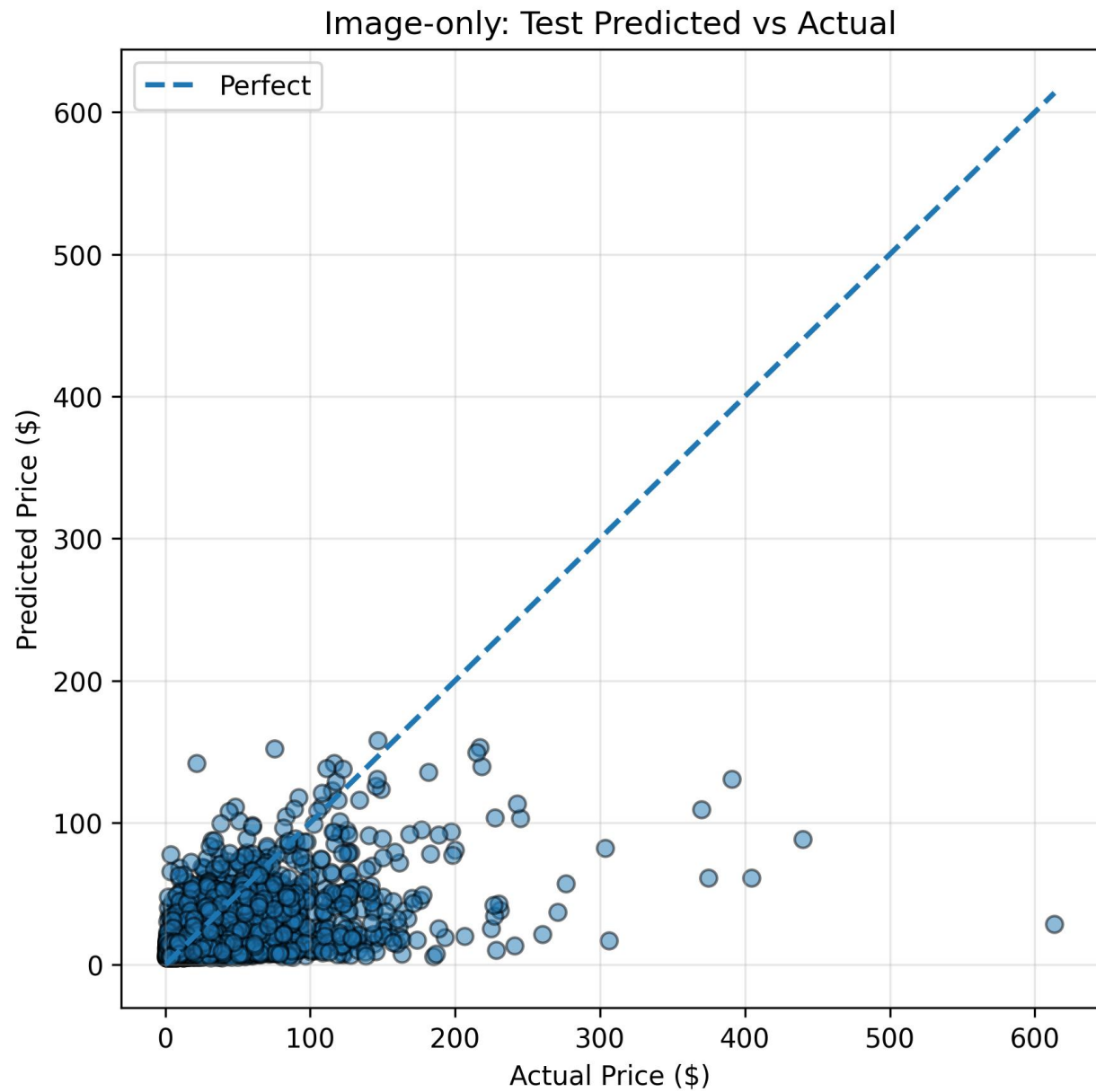
Potamianos, G., Neti, C., Gravier, G., Garg, A., & Senior, A. (2003). Recent advances in the automatic recognition of audiovisual speech. *Proceedings of the IEEE*, 91(9), 1306–1326.

<https://doi.org/10.1109/JPROC.2003.817150>

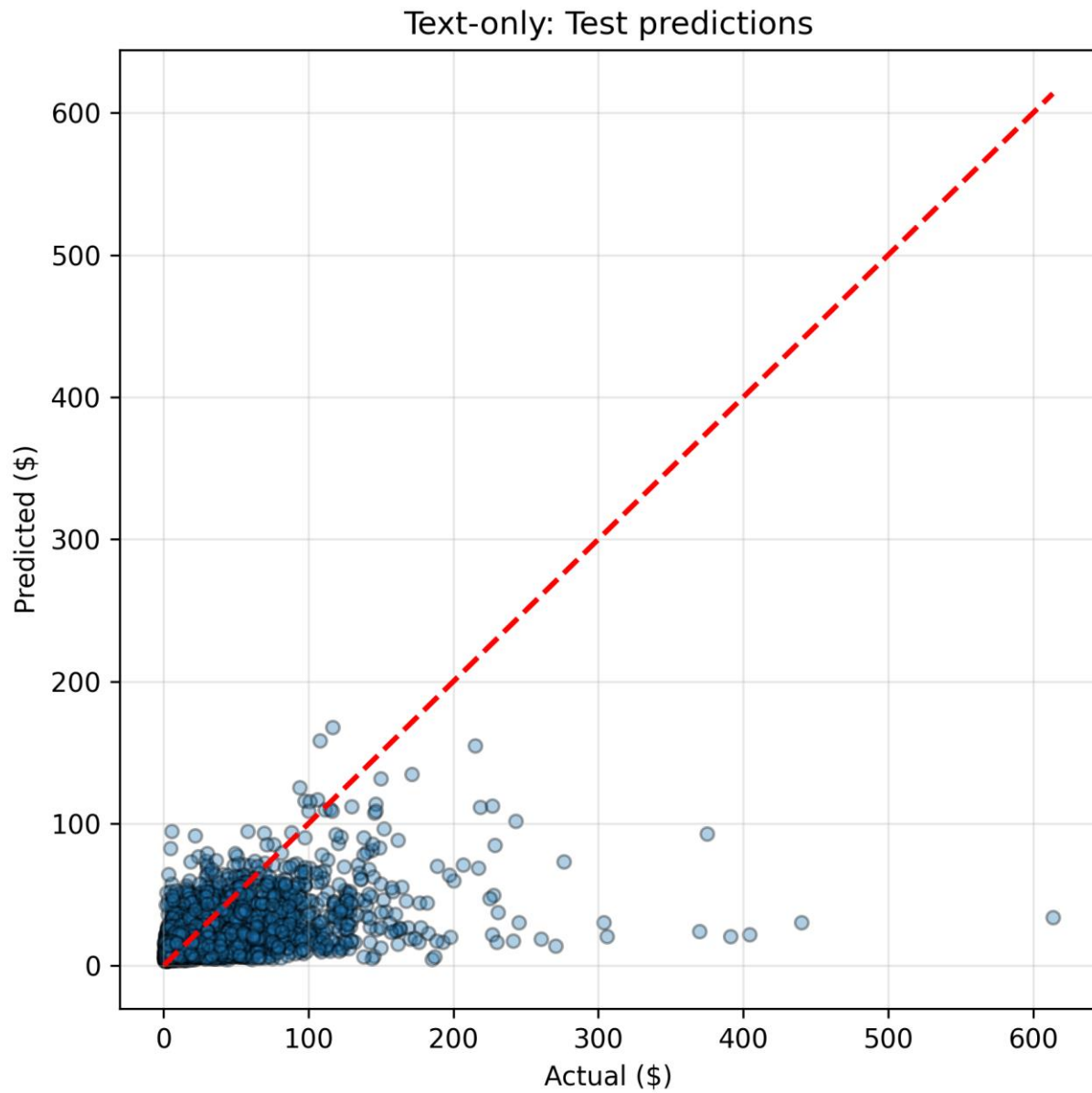
Yue, X., Zhang, Y., Hu, S., Lin, T., Chen, T., Wang, L., ... Zhang, D. (2024). *MMMU: A massive multi-discipline multimodal understanding benchmark*. arXiv:2401.11102.

<https://arxiv.org/abs/2401.11102>

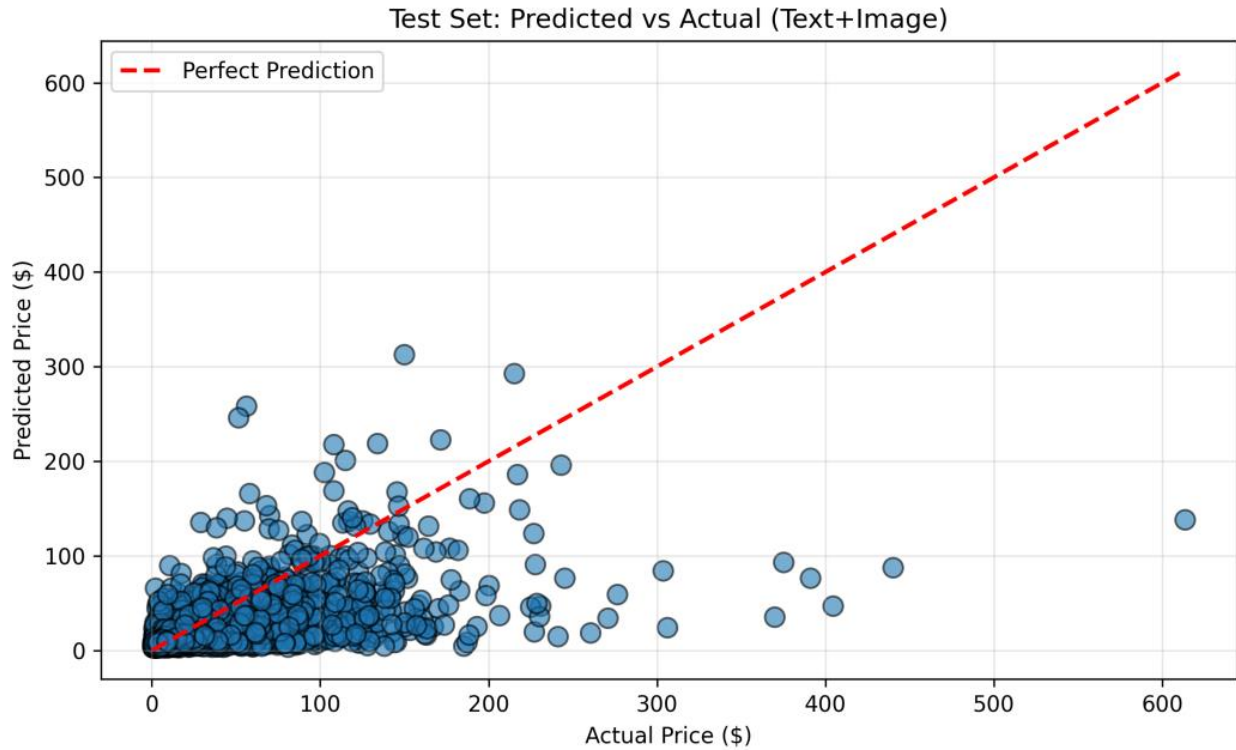
Appendix



Appendix 1. True vs Predicted Price Image Only.



Appendix 2. True vs Predicted Price Text Only.



Appendix 3. True vs Predicted Price Combined Model.

Appendix 4. `appendix_asset.txt` in the submission contains raw data for the ten worst performing predictions.